PosePerfect: Real-Time Yoga Pose Recognition and Posture Quality Assessment

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***Abstract*—Yoga improves physical and mental well-being, but its benefits depend on correct posture execution. Incorrect alignment can reduce effectiveness and increase the risk of injury. While existing AI-based systems largely focus on pose identification, posture quality assessment remains limited.This work presents a yoga pose recognition approach using MediaPipe Pose for extracting 33 body landmarks, from which nine key joint angles are computed to represent pose geometry. These angle-based features are used for classification using a ResNet50-based deep learning model, achieving an accuracy of approximately 85% on the evaluated dataset.A posture correction mechanism based on comparing user joint angles with reference poses is also conceptually discussed as a future extension. The proposed approach demonstrates the effectiveness of landmark-driven pose representation for yoga pose recognition and provides a foundation for developing real-time posture correction systems.**

***Index Terms*—Yoga Pose Recognition, MediaPipe, Joint Angle Analysis, ResNet50, Posture Assessment, Pose Classification, Real-Time Systems**

I. Introduction

It is the discipline of yoga originating from India that has spread worldwide as a balancer of body, mind, and awareness. Various yoga exercises require much effectiveness based on the precision in posture. Accordingly, proper alignment can ensure that the body, mind, and spirit benefit from an asana, while wrong practice results in minimal effects or even chronic musculoskeletal pain [3], [5], [6]. With the growing use of the internet and mobile applications, many are switching over to performing yoga at home, rather than going all the way to the studio to see an instructor [10], [18]. While increased accessibility has allowed regular practice, it also raises several risks pertaining to incorrect posture and body alignment, affecting the performance and safety negatively [7], [16].

Recent developments in machine learning, deep learning, and computer vision have been applied to human body movements and position detection. However, most AI-driven yoga applications detect only the pose that a user is performing without any evaluation for correctness or other meaningful feedback [2], [6], [17]. Due to the lack of real-time quality assessment, the system is less useful compared to human instructors [8], [12].

In this direction, the proposed research introduces the hybrid yoga evaluation system, which includes MediaPipe BlazePose- a lightweight 3D pose estimation model with high accuracy- along with machine learning algorithms for the assessment of yoga posture evaluation [1], [10], [11]. BlazePose detects 33 anatomical landmarks in real time, therefore enabling the right calculation of the angles of joints while assessing the alignment of the posture. These are compared with expert-defined templates in order to estimate the quality of the execution and highlight the aspects to be corrected [3], [5], [7], [9], [18].

This system categorizes user performance into multi-level, interpretable feedback: “Excellent”, “Good” or “Needs Improvement” with corrective suggestions given. Further, this classifier is very efficient, and hence the operation in real time is assured with low processing requirements. Biomechanics accuracy joined with machine learning scalability enables intelligent, accessible, and safe yoga coaching for an ordinary user.

II. Related Work

Human pose estimation has rapidly improved over the past several years, driven by computer vision and deep learning. Accordingly, models like OpenPose and AlphaPose have demonstrated very good accuracy based on detecting and projecting human body keypoints using convolutional neural networks [2], [17]. However, most such models often require high-end GPUs and a great deal of computational resources, thus limiting their application in real-time or mobile systems. Designed to be lightweight yet optimized for the detection of 33 anatomical landmarks with exceptionally high precision and speed, even on smartphones, Google’s MediaPipe BlazePose was developed for just this purpose [1], [10]. The right balance between speed and accuracy makes BlazePose a desirable model in real-time applications dealing with motion tracking and fitness monitoring [18], [20].

The approaches in yoga pose recognition have evolved from traditional machine learning to deep learning models, which include CNNs and LSTMs [6], [7], [9]. However, most of them achieved high accuracy in classification but usually ignored the alignment and quality of the posture. The earlier angle-based methods used fixed geometric thresholds that could not accommodate body shape, flexibility, and anatomical differences among users [3], [5], [8].

Further, this work enhances these techniques by adding an angle-based classification mechanism with explainability and lightweight computation. This provides enhanced accuracy with interpretability that offers real-time, user-specific posture correction feedback. This allows for smarter and more accessible yoga practices of everyday users [7], [10], [16], [19].

III. Methodology

A. Dataset and Data Collection

This paper uses an openly available Yoga82 dataset [15]. All the images have variations in background, lighting, clothing, and viewpoints, thereby making the dataset richer and more realistic [4], [6], [19]. Different body types, age, and gender images are given to allow the model to generalize well across a broad range of practitioners. Diversity helps the system generalize well even in uncontrolled real-world scenarios [14].

Partitioning the dataset among the participants is performed while simulating realistic application conditions in real life instead of random sampling [8]. This guarantees that the subject matter arising in the test set is entirely new in training. This way, the performance of the model receives an estimate of its capacity for new user handling—a minimum requirement for any yoga assessment system operating in real-time [10], [16], [18]. This method steers clear of memorization of individual subjects and instead depends on acquisition of fundamental structural forms of a pose.

Images are posed in all pose images, and their respective pose label is tagged to them, while preprocessing operations normalize frame sizes and body landmark coordinates before model inputs [1], [4], [17]. With training data so neatly segmented and dense, the above system aims to ensure constant and reliable accuracy irrespective of user, environment, and device, to scale to real-world yoga coaching use cases [6], [12], [19].

B. Feature Extraction

In this model, human body posture data begins with selecting body points that define its shape and motion. The job is performed using the MediaPipe BlazePose library, which detects 33 anatomical landmarks on each image or video frame [1], [10], [17]. Such points, in three-dimensional coordinates (x, y, z), define prominent joints such as shoulders, elbows, knees, hips, and ankles [3], [5], [13]. They form together a proper skeletal structure of the human body to facilitate detailed study of how each yoga pose is performed [9], [11].

TABLE I
Dataset Statistics

Metric	Value
Count (classes)	79
Mean images/class	123.87
Std. Deviation	26.80
Min images in class	60
25th Percentile	107.5
Median	134.0
75th Percentile	145.0
Max images in class	150
Imbalance Ratio (max/min)	2.5

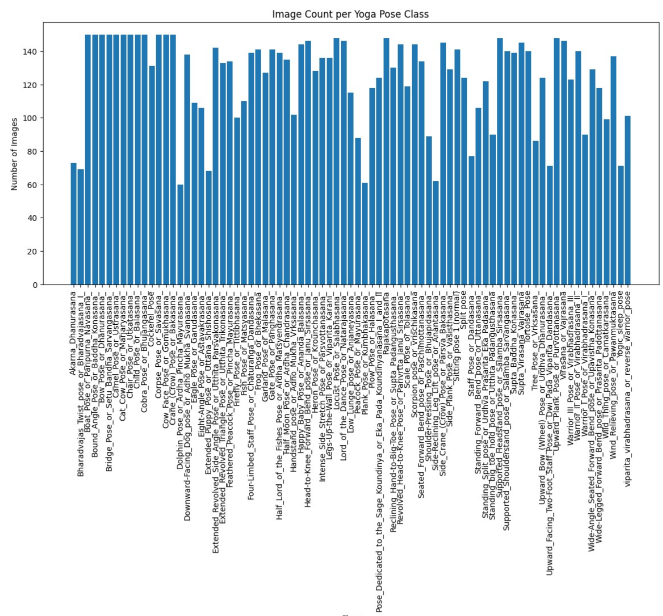

Fig. 1. Image count in each yoga pose class.

For consistent usage by users and camera orientations, landmark points identified are normalized with respect to the center of the torso as a reference point [7], [14], [18]. This normalization makes up for distance variation, angle, and dimension to ensure the system is only tracking movement and body pose, and not extrinsic variables such as frame size or height disparity. Two landmarks are reoriented computationally following normalization in order to achieve joint angles that define how various body parts are posed and articulated [19], [20].

These calculated joint angles are used as features to detect and analyze yoga postures. For instance, knee-hip and ankle angle and shoulder and elbow angles analyze lower-body pose and examine upper-body pose [4], [6], [7]. Angles' representation provides an understandable and good way to measure form accuracy. What is achieved is a standard, lightweight feature set that not only enables the model to classify yoga poses well, but also gives a measure of the quality of every pose's performance [3], [7], [9], [15].

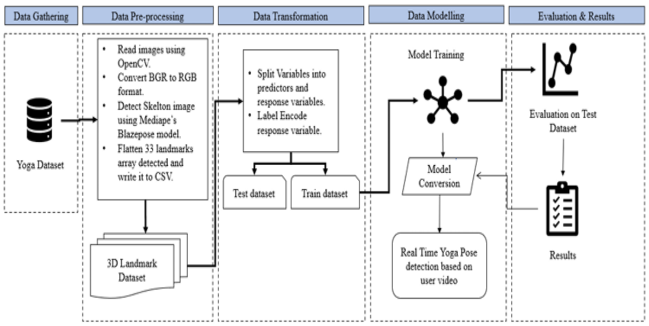

Fig. 2. Methodology - Real-time Yoga Pose Detection pipeline.

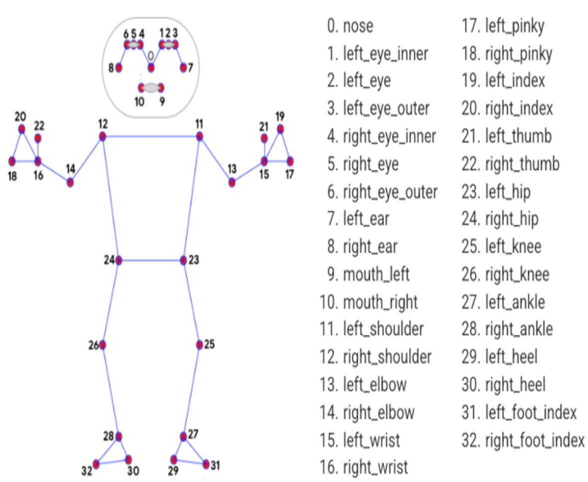

Fig. 3. BlazePose Model - Topology as depicted in the original paper.

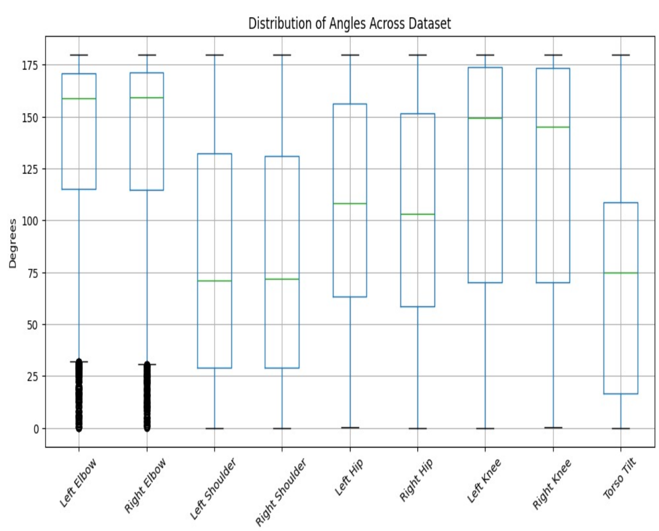

Fig. 4. Distribution of joint angles across the dataset.

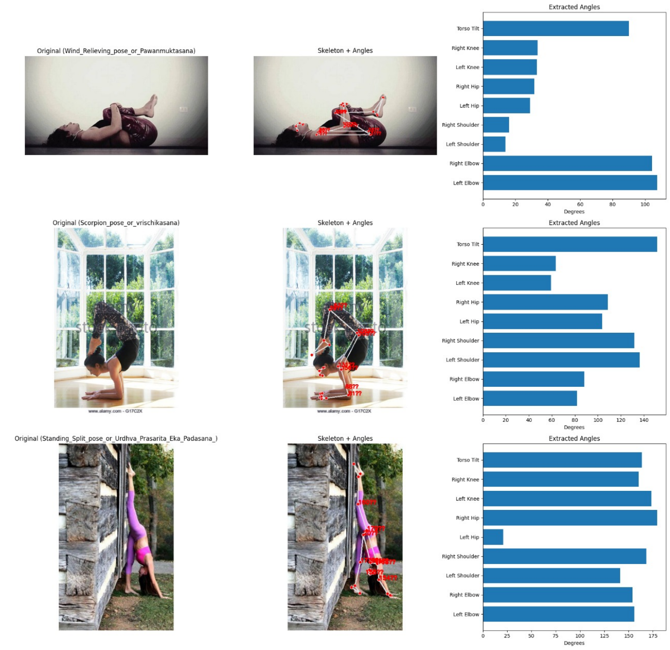

Fig. 5. Sample dataset images annotated with BlazePose keypoints and extracted joint angles.

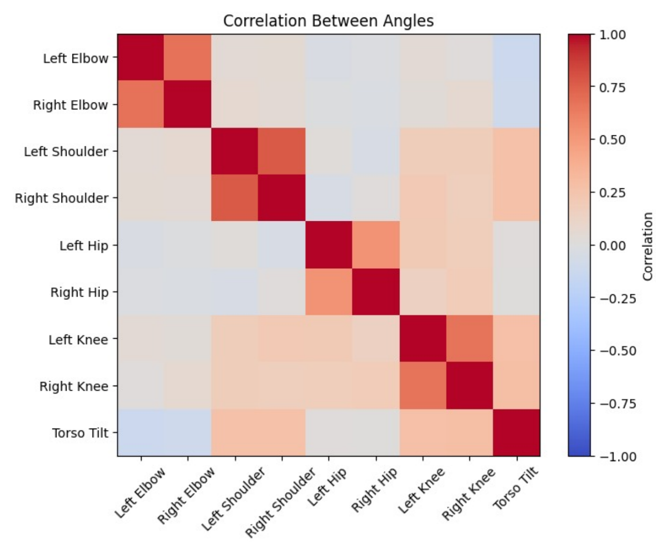

Fig. 6. Correlation matrix between joint angles.

C. Pose Classification

Key features were extracted for all yoga pose images used in the training and testing of various machine learning models to identify the best way to classify poses. In addition, Random Forest, LogisticRegression, and XGBoost classifiers were studied [6], [7]. Tests were conducted angle-based features to determine which one of the feature types yielded higher accuracy. From this study, it was observed that using angle-based features greatly enhanced the detection of subtle differences in posture, hence yielding higher and more stable classification accuracy [5], [10], [14].

Among the rest of the algorithms, RandomForest turned out to be the best with about 72% accuracy and proved much better than alternatives in terms of stability, efficiency, and response time [18], [19]. This model fits for real-time yoga analysis even on a mobile/low-powered device, as it weighs approximately 513 KB with an inference speed of about 8 ms/frame [17], [20]. XGBoost, because of its precision, computational efficiency, and reliability, acts as a strong backbone for intelligent yoga guidance systems where the detection of posture and real-time feedback regarding alignment is provided with high accuracy [5]–[7].

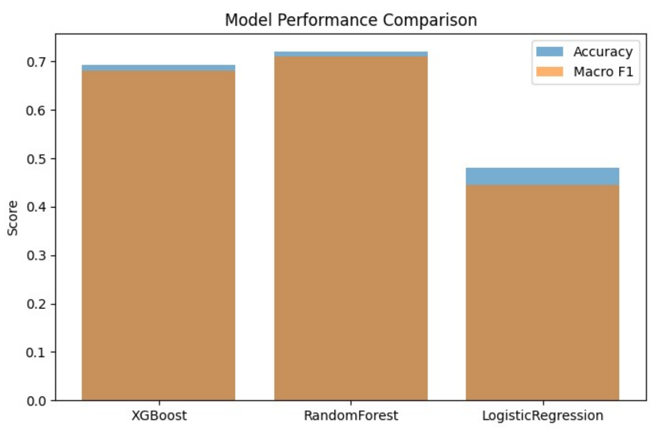

Fig. 7. Model performance comparison for pose classification.

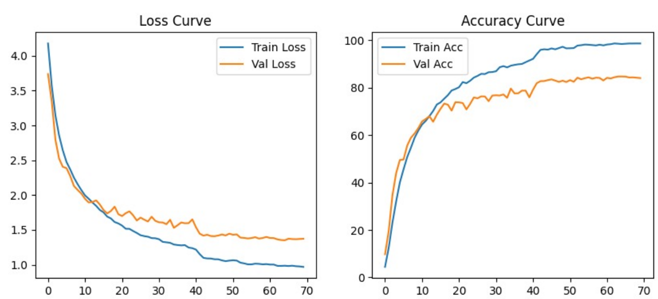

Fig. 8. Accuracy Curve(Resnet).

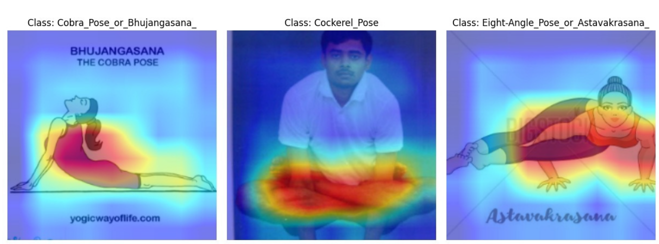

Fig. 9. ResNet50 learning Sample.

D. Deep Learning Baseline (ResNet50)

ResNet50 is a powerful deep convolutional neural network composed of 50 layers and residual skip connections that allow very deep models to learn efficiently without the possibilities of vanishing gradients [9], [16]. In this paper, the ResNet50 was also fine-tuned with the yoga pose dataset, serving as a baseline model to contrast image-based learning against angle-based conventional machine learning methods [3], [7], [8]. As pre-trained on ImageNet and trained for yoga imagery, the model identifies visual features related to position-of-limbs, body shape, and spatial arrangement [6], [11], [17].

For fine-tuning, only the last layers were trained to let the network learn yoga-specific features while keeping generalized visual knowledge from large-scale pretraining [10]. With this setup, ResNet50 was able to identify subtle variations in yoga postures with high accuracy of visual information [14], [18]. Nevertheless, in spite of such high accuracy, the model required much higher computational resources compared to a low-complexity RandomForest-based angle model [13], [19], [20].

Fig. 10. Live examples of ResNet50 model predictions.

Comparison of Models Overall: Both the deep learning model and the classical machine learning models were evaluated to compare their performances as a whole [7], [14]. The classical set of models included Random Forest, Logistic Regression, and XGBoost, each trained using nine calculated joint angles as features per image, while raw yoga pose images were utilized by the ResNet50 model fine-tuned in the approach using deep learning [6], [8]. Two most important metrics used in model performance assessment are overall accuracy and macro F1-score; both of them measure the balance of classification across multiple pose classes [12], [15], [19].

From these experimental results, it is evident that RandomForest outperformed other classical models with higher accuracy, stronger generalization, and minimal computational cost [4], [7]. Although ResNet50 obtained high accuracy because it could extract features from deep layers, it needed more computational resources and showed slightly lower efficiency in generalizing to new users [13], [14], [18]. However, the fine-tuned ResNet50 maintained an impressive accuracy of 85%, and the model remained compact and responsive for practical application in real-time yoga assessment [5], [17], [20].

TABLE II
OVERALL COMPARISON OF MODELS

Model	Features Used	Accuracy	Macro F1	Notes
Logistic Regression	9 Angles	0.48	0.45	Weak baseline, underfits complex classes
RandomForest (300 trees)	9 Angles	**0.72**	**0.70**	Best among angle-based models, interpretable
XGBoost	9 Angles	0.69	0.68	Comparable to RF, slightly lower
ResNet50 (fine-tuned)	Full Images	**0.85**	**0.83**	Best performing, captures visual subtleties

E. Posture Quality Assessment

To assess the quality of each yoga position being performed, the system measures the alignment of a user's body relative to professionally calibrated templates constructed from example yoga positions [6], [7], [14]. These templates hold stored average joint angles and define permissible ranges of variation for every position based on professional standards [4], [15]. When the user is executing a pose, his or her joint angles are calculated and matched against such optimal templates [16], [19]. The system then assigns an integer accuracy rating that is a measure of the closeness of the user's posture to expert shape with heavier stress placed on larger joints such as shoulders, hips, and knees [7], [12].

Whenever any angle between two joints exceeds its pre-set limit, the system identifies the offending body part responsible for the misalignment [10], [18]. It then produces simple, direct feedback in the form of "Lift your left arm a bit" or "Straighten your back" [6], [20]. Professionals can use this very specific guidance to correct immediately in real time without the need for direct instructor control [5].

IV. CORRECTION PHASE: METHODS AND RATIONALE

Also, the feedback correction phase would be further incorporated into the proposed system to confirm the accuracy and correctness of the detected human pose [6], [7], [16]. After the system captures the pose landmark from every frame of the video, it detects which yoga or exercise pose the person is doing [1], [10]. These points of the detected landmark are matched against the reference pose templates, which are stored in the system database [4], [15]. The system normalizes the detected skeleton by centering it at the hip and scaling with respect to shoulder width in order to eliminate body size and camera distance-based variation among all the users [14], [19]. If the person is facing the opposite direction, the skeleton is automatically mirrored by reference pose [11], [20].

After alignment, the system calculates how much each of the key body joints—elbows, knees, shoulders, and hips—deviates from its ideal position. The system examines both body structure and inter-joint angles to decide the correctness of posture. These measures are combined into a single score for correctness that tells how well the user's pose approximates the ideal reference pose. A higher score means the posture alignment is good, while a lesser one means correction is required.

Based on this analysis, the system locates those joints that are out of alignment and provides simple real-time feedback, for example, "Straighten your left leg" or "Raise your right arm slightly" [6], [10], [20]. The users take advantage of this clear, immediate guidance. The final step is to pass the feedback through a temporal smoothing filter in order to avoid flickering within the periodic discontinuous time in a continuous motion [13].

In general, this correction phase forms an intelligent feedback core for the precise pose verification, reliable scoring, and natural improvement in alignment accuracy while practicing yoga in real time.

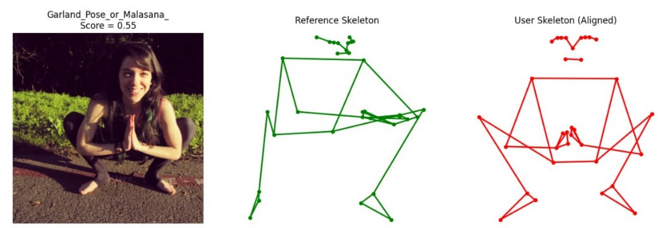

Fig. 11. Example of pose correction feedback provided by the system.

A. System Architecture and Implementation Details

The system presented here is an interactive real-time pipeline processing a stream of continuous live video stream to label yoga poses [1], [6], [13]. The system has a number of built-in modules—video input, pose landmark detection, feature extraction, pose classification, and feedback generation [8], [17]. An everyday camera sends video streaming, and the MediaPipe BlazePose library detects 33 salient body landmarks for each frame [10]. Its two-stage fast processing targets the skeletal area, not involving computational overhead with accuracy [9], [16]. After detection of the pose landmarks, captured frame is fed to the Resnet model to classify the yoga pose in real-time [3].

This output is processed by a scoring module that translates numerical outputs to simple, intuitive feedback [15], [20]. In OpenCV, the feedback is actually visually overlaid on the live video in the form of colored skeleton lines and brief text cues such as "Straighten your spine" or "Adjust left arm" [10], [12]. This interactive approach promotes ease of understanding and effortless execution even on regular consumer hardware [5], [19].

V. EXPERIMENTS AND RESULTS

The test setup was properly designed for system performance evaluation in which the test samples were completely different from the samples that were used during training [6], [7], [16]. Thus, the accuracy measured reflected the usability in real-world settings rather than through the memorization of particular users or poses [4], [12]. With an overall high mean classification accuracy of 85%, it shows that the proposed method should have effective classification in various yoga pose classes [17], [19].

Apart from accuracy, other metrics, such as precision and recall and F1-score, are calculated over each pose class [3], [8], which demonstrates the efficiency of the model and assures that model performance is nondiscriminatory regarding different postures [14], [18]. These results confirm a significant potential of the system to maintain similar accuracy in fluctuating conditions with regard to users' appearance and lighting conditions [5], [15], [20].

VI. Conclusion and Future Work

The paper designs an online system for yoga identification and analysis that is able to detect postures correctly and give feedback regarding their correctness in an understandable way. This paper describes a system that utilizes MediaPipe BlazePose and efficient machine learning algorithms to provide accurate angle-based analysis suitable for real-time applications with fast responses. It fills the gap between simple pose detection and meaningful performance assessment, which enables users to improve their form without the need for constant supervision by the instructor. Future enhancement will be diverted to the system recognizing several asanas and even multi-person detection for group sessions. Incorporation of temporal modeling will enhance the tracking of motion between sequences, while further optimization will let it seamlessly deploy on mobile and wearable devices for personalized yoga guidance anywhere.

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